

EDA Practice Task Sheet

Dataset: test_eda_practice.csv **Difficulty:** Beginner → Intermediate

Goal: Perform complete Exploratory Data Analysis and discover the hidden data-quality issues.

Part 1 — Load & Understand the Data

- Q1. Load the CSV using Pandas. Read test_eda_practice.csv into a DataFrame called df.
- Q2. Find the number of rows, number of columns, first 5 rows, last 5 rows, and column names.
- Q3. Display the data types of every column. Identify numerical, categorical, and date/timestamp columns.
- Q4. Use .info(). Identify columns with missing values, suspicious data types, and columns with many unique values.

Part 2 — Data Quality

- Q5. Find the number and percentage of missing values per column. Which columns require attention?
- Q6. Check for duplicate rows. Find the number and percentage of duplicates.
- Q7. Investigate unique values for gender, category, and state. Look for suspicious or inconsistent values.
- Q8. Check whether categorical values are consistently formatted. Look for differences in capitalization and unexpected spaces. Discover the problematic values yourself.

Part 3 — Numerical Analysis

- Q9. Use df.describe() and analyze amt, city_pop, lat, long, merch_lat, and merch_long. Identify mean, median, minimum, maximum, and standard deviation.
- Q10. Analyze amt. Create a histogram and boxplot. Determine whether it is skewed and whether potential outliers exist.
- Q11. Use the IQR method to identify outliers in amt. Calculate Q1, Q3, IQR, lower bound, upper bound, and number of outliers.

Part 4 — Categorical Analysis

- Q12. Analyze gender. Create value counts and a bar chart. Calculate percentages and investigate missing/unusual categories.
- Q13. Analyze category. Create a bar chart. Identify categories with the most and least transactions and investigate inconsistent names.
- Q14. Find the Top 10 merchants based on transaction count and visualize them.

Part 5 — Fraud Analysis

- Q15. Analyze is_fraud. Find the number and percentage of 0 and 1 values. Determine whether the target is balanced or imbalanced.
- Q16. Calculate fraudulent transactions and fraud rate for each category. Compare the observed fraud rates.
- Q17. Compare amt for is_fraud = 0 and is_fraud = 1. Calculate mean, median, minimum, and maximum and create a suitable visualization.
- Q18. Analyze fraud rate by gender and create a suitable chart.

Q19. Find the Top 10 states by fraud rate and also report transaction volume. Consider whether low-volume states should be interpreted the same way as high-volume states.

Part 6 — Date & Time Analysis

Q20. Convert `trans_date_trans_time` to a proper Pandas datetime type. Create year, month, day, hour, and day-of-week features.

Q21. Find transaction counts for every hour of the day and create a visualization.

Q22. Calculate fraud rate by hour. Identify hours with higher and lower observed fraud rates.

Q23. Analyze transaction volume and fraud rate by day of week.

Part 7 — Customer Analysis

Q24. Use `dob` to calculate customer age. Find minimum, maximum, average, and median age.

Q25. Investigate age data quality. Look for negative or extremely high ages and count affected records. Do not clean them yet.

Q26. Create age groups: <20, 20–29, 30–39, 40–49, 50–59, and 60+. Analyze transaction count, fraud count, and fraud rate.

Part 8 — Correlation & Relationships

Q27. Create a correlation matrix and heatmap for numerical variables. Identify strong and weak correlations.

Q28. Create a visualization comparing `amt` and `is_fraud`. Describe observed patterns.

Q29. Investigate the apparent relationship between `city_pop` and `amt`.

Part 9 — Advanced EDA Challenges

Q30. Find high-value transactions. Choose a threshold based on your EDA. Analyze count, fraud percentage, categories, gender, and state.

Q31. Build a summary of fraudulent transactions using category, amount, gender, state, city population, hour, and day of week. Identify patterns without claiming correlation proves causation.

Q32. Find merchants with the highest number of fraudulent transactions. Compare fraud count with total transaction volume. Explain why fraud count alone can be misleading.

Part 10 — Data Cleaning Challenge

Q33. Identify columns containing missing values.

Q34. Identify duplicate records.

Q35. Identify inconsistent categorical values.

Q36. Identify numerical columns containing potential outliers.

Q37. Identify invalid values.

Q38. Decide which columns should potentially be dropped and give a reason for each.

Q39. Identify columns requiring transformation, such as `datetime` → `datetime` features, `categorical` → `encoded` values, and `DOB` → `age`.

Final EDA Report

Write a short report covering: Dataset Overview, Data Quality, Customer Insights, Transaction Insights, Fraud Insights, and Final Observations.

Include 5–10 observations from your EDA. Use careful wording: describe observed patterns rather than claiming that a variable causes fraud.

Bonus Challenge

Try to complete the entire EDA using only Pandas, NumPy, Matplotlib, and Seaborn.

Do not look for the intentional problems beforehand. The goal is to discover them through your own analysis.